

You will need to submit a write-up for your final project. The format should be loosely structured in the format of a scientific paper or a data analysis report. The structure is simply for guidance. While the product should be coherent for providing information like basic background on your question and discussion of your results, the grade will be primarily based on your Methods section.

The Methods section should include a comprehensive discussion of the data that you used (source, size, format) and the breadth and comparison of techniques you researched and employed (data structures, algorithms and technologies) to proficiently work with your data. The focus should be on skills you learned during this second half of the course. In particular, I will be focusing on how you treat the following areas:

Structure of your submission (2 credits) – The structure of your submission should loosely follow the format of a scientific paper or a data analysis report. Keep the main body succinct within 6-8 pages and use Appendix (no page limitation) for supporting tables/charts.

Complexity of data (5 credits) – You are encouraged to pick a complex data and demonstrate your effort to process the data using what you learned from class. A partial finished work on a complex data may get the same or more credit as the one of a fully finished work on a significantly simpler data as long as you demonstrate the following:

- a) sufficient assessment of the data complexity that illustrate the difficulties you experience, and
- b) preliminary results that demonstrate your code capability on a small portion of your sample data, and
- c) list challenges you face and show us the reasons why your code should work on the full sized dataset once you resolve the challenges

On the other hand, if your data is significantly simpler that is not able to demonstrate your ability to apply knowledges we learn at class, you will lose not only credit in this grading component, but also credits in the following grading components (listed below). Hence, choose your data carefully.

Data processing technologies (3 credits) – tidy data; storage consideration / mechanisms for storing your data; or, use of MapReduce or Spark to process your data.

Data manipulation (2 credits) – use of grammar for data manipulation or SQL.

Theoretical evaluation and visualization (3 credits) - Algorithms used/not used, and a discussion regarding the reason for your choices; if using a custom algorithm, then an evaluation of the algorithm used; and, data structures you may have used to represent your data, visualization of your results including time-series plots, scatter plots, histograms, barcharts etc

Peer evaluation (5 credits) – Credits given by your teammates anonymously. This provides a structured learning process for students to contribute to the group project collaboratively.

For each area (except the peer evaluation), consider implementing using multiple methods and compare/contrast the methods you used. If you choose to implement using one single method, make sure to document your reasoning of why you choose this method over the other. Use theories you learned at class and come up with empirical evidence to support your proposed method.

Jupyter Notebook submission

You must submit code that can **run** and **recreate** your results in Docker or Google Cloud. Failure to do so will result in a significant grade deduction in your final project grade. It should be well-documented and contain useful descriptive inline code comments. Variable and function names should be selected with an emphasis on code legibility. What's more, it's important to submit your notebook with a fresh re-run to ensure the codes in your notebook are executed in the right sequence.

Document the environment you used to run your Jupyter Notebook. For example, if you choose to run your code in docker, specific which docker container (name, version, date etc) you used and list steps you used to create docker. For Google Colab and other environments, submit the "pip list" output with your notebook.